# TAISIIA MAMENKO

**SOFTWARE ENGINEERING STUDENT | PYTHON | MODERN C++ | QT6 UI & DATA TOOLS**
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Engineering Science student who builds tools people use: Qt6 interfaces, Python APIs (FastAPI, Flask) and command-line tools, and data validation that catches bad records early. Writes clear, tested code with CI on Linux and Git pull requests.

## EDUCATION

**Simon Fraser University | Bachelor of Applied Science, Engineering Science** **Expected Graduation: May 2029**
CGPA: 3.87 | Burnaby, BC Available full-time May-Dec 2027

## TECHNICAL SKILLS

**Languages:** Python, C++ (C++17/20), SQL, Java, MATLAB, Bash
**UI, APIs & Tools:** Qt6 Widgets, custom QPainter widgets, FastAPI, Pydantic, Flask, Grafana, command-line tools
**Software Design:** Object-oriented design (C++, Python dataclasses, SQLAlchemy ORM), RAII, layered architecture, data validation
**Testing & Debugging:** GoogleTest, CTest, pytest, coverage, AddressSanitizer, UBSan, mypy, Ruff, strict warnings
**Git & Linux:** Git/GitHub (feature branches, pull requests), GitHub Actions CI, Linux (Ubuntu), Bash, systemd, Nginx, Docker
**Data:** SQLite, PostgreSQL, SQLAlchemy 2.0, CSV and JSON processing, Prometheus

## PROJECTS

**BuildScope AI** | Python, FastAPI, SQLAlchemy, SQLite, C++20, CMake **Oct 2026 (In Progress)**
*github.com/liltaya/buildscope-ai | Solo project: local build-pipeline reliability tool*

- Built a FastAPI service that registers local code repositories and auto-detects each one's language, build system (setuptools, Poetry, Hatchling, Flit, CMake), and test framework (pytest, GoogleTest, CTest), with the evidence behind every result.
- Designed a layered Python architecture (API routes, Pydantic schemas, service, store, SQLAlchemy models) with typed domain errors mapped to 404, 409, and 422 responses; path validation, unique and check constraints, and duplicate detection keep bad records out.
- Wrote 70 pytest tests (unit, API, integration) with Ruff and mypy CI gates; started a C++20 worker library with GoogleTest CI.

**Construction Analytics Dashboard** | C++20, Qt6, SQLite, CMake, GoogleTest **Sep-Oct 2026**
*github.com/liltaya/Construction-dashboard | Solo project: designed, built, tested, and documented every component*

- Built a Qt6 desktop tool (about 4,600 lines of layered C++20: UI, service, analytics, repository, RAII-managed SQLite) that turns raw project data into four interactive views so users can spot budget overruns and overdue work.
- Built the UI with a project selector, live search, category and status filters, sortable tables, and a custom QPainter chart widget with hand-written drawing, scaling, and layout math instead of a charting library.
- Designed a CSV import path that catches data integrity issues before they reach the database: header, field-count, ID, status, and budget checks with row-numbered error messages, plus all-or-nothing SQLite transactions that roll back on failure.
- Wrote 25 GoogleTest tests and a two-job GitHub Actions pipeline that builds headless on Ubuntu with strict warnings and reruns every test under AddressSanitizer and UBSan; documented the architecture and data model.

**Operational Data Platform & Log Analyzer** | Python, Flask, SQLAlchemy, PostgreSQL, Docker, Linux **Jul-Aug 2026**
*github.com/liltaya/Flask-service | Solo project*

- Built a Python REST API (Flask, SQLAlchemy, PostgreSQL) with input validation, structured errors, health checks, and Prometheus metrics feeding Grafana dashboards; deployed on Linux with Nginx, systemd, and Docker Compose.
- Wrote a Python command-line log analyzer (dataclasses, type hints, text/JSON/CSV reports) that flags error rates, malformed lines, and suspicious requests (path traversal, SQL injection, scanners); benchmarked on 1M generated records.
- Wrote 38 pytest tests and a four-job CI pipeline (Ruff, mypy, tests, coverage); fixed failing CI checks through pull requests.

**Virus.exe Card Game** | C++, Object-Oriented Design **Fall 2026 (In Progress)**
*SFU team course project | My role: game concept, rules, and class design*

- Proposed the concept to my team and designed the rules for a 2-player strategy card game in which each player defends a computer system (CPU, Memory, Network, Storage); the first to destroy 3 of the opponent's 4 systems wins.
- Designed the C++ object model (Card, Deck, Player, ComputerSystem) that maps a standard 52-card deck to attacks, repairs, and face-card abilities such as firewall bypass, antivirus block, and zero-day attacks.

**Engineering Coursework & Research** | Simon Fraser University **Sep 2025-May 2026**

- **Boeing 737 Software Assurance Research** (Sep-Dec 2025): report on how safety-critical aircraft software is verified and validated.
- **MATLAB Engineering Analysis** (Jan-May 2026): modeled engineering scenarios, automated calculations, and validated results.

## EXPERIENCE

**PlotUp Inc.** | **Branding Specialist** | Burnaby, BC **Jan-Jul 2025**

- Owned the company's visual identity end to end: designed the logo and business cards and built the Wix website.
- Turned stakeholder feedback into user-facing website and content updates, coordinated revisions, and delivered on schedule.

## GAMES & INTERESTS

- Built a small game for children who struggle with concentration; watched players use it, noted what confused them, and iterated.
- Grew up playing EA SPORTS FIFA and The Sims, the start of a long-running curiosity about how games are built.